Tejaswini Katale

+1 713-842-2375 tkatale@cougarnet.uh.edu Portfolio Scholar LinkedIn

EDUCATION

University of Houston — **Aug 2025 – May 2027**
*Master of Science in Computer Science (Thesis) - **3.61**/4* — *Texas, USA*

Vishwakarma Institute of Technology, Pune — **Aug 2020 – May 2024**
*Bachelor of Technology in Computer Engineering - **8.65**/10* — *Pune, India*

Selected Publications

Gasoline Prices and Transit Ridership: Evidence from 50 U.S. Transit Agencies, 2002–2026 *(Under Review)*
Bo Chen, ***Tejaswini Katale***, Lu Gao
Transportation Research Record (TRR)

Pavement Texture Distributions and Wet-Weather Roadway-Departure Crashes *(Accepted for Presentation)*
Nasir Gharaibeh, Yussuf Tarnini, Mohammad Talebzadeh, Boni Kutela, Jingran Sun, ***Tejaswini Katale***
Transportation Research Board (TRB) 2027 Annual Meeting

Exploring OSHA Construction Accident Records with Multimodal AI Models
Islem Sahraoui, ***Tejaswini Katale***, Gowtham Reddy Gudla, Lu Gao, Kinam Kim, Zia Din, Ahmed Senouci
Construction Research Congress, ASCE, 2026

Developing an AI Assistant for Knowledge Management and Workforce Training in State DOTs
Divija Amaram, Lu Gao, Gowtham Reddy Gudla, ***Tejaswini Katale***
Electronics, Special Issue: Artificial Intelligence-Driven Emerging Applications, 2026

LLM-Guided Hybrid Simulation for Airport Cyber-Resilience Assessment
Tejaswini Katale, Lu Gao, Yongxin Liu, Dahai Liu, Hongyun Chen
Mathematics, Special Issue: Mathematical Methods in System Engineering Modeling and Simulation, 2026

Enhancing Pavement Safety in Texas: Crash Susceptibility Modeling, GIS Tool, and Hydroplaning Risk Assessment
Nasir G. Gharaibeh, Lu Gao, Boniphace Kutela, Yussuf Tarnini, Mohammad Talebzadeh, ***Tejaswini Katale***
Texas A&M Transportation Institute, TxDOT Technical Report, 2026

Integrating Pavement Condition Records with LLM-Based Crash Narrative Analysis for Pavement Safety Assessment
Sarayu Varma Gottimukkala, Lu Gao, Nasir Gharaibeh, Yussuf Tarnini, Mohammad Talebzadeh, Boni Kutela, ***Tejaswini Katale***
Accident Analysis & Prevention, 2026

A Bilevel Optimization Framework for Adversarial Control of Gas Pipeline Operations
Tejaswini Katale, Lu Gao, Yunpeng Zhang, Alaa Senouci
Actuators, 2025

Recession Forecasting: Domain-Specific Feature Selection Methodology *(Best Paper Award)*
Tejaswini Katale, Jayashri Hase, Aakash Chotrani, Sakshi Ozarde
2024 First International Conference for Women in Computing (InCoWoCo)

POSTER PRESENTATIONS

Domain-Adapted AI for Pavement Safety Analysis
Tejaswini Katale, Lu Gao, Nasir Gharaibeh
International Conference on Transportation & Development (ICTD), 2026

Large Language Models in Pavement Safety Analysis: A Comparative Study of Open-Weight Models
Tejaswini Katale, Gowtham Gudla Reddy, Lu Gao, Nasir Gharaibeh
ASCE Transportation Conference, 2026

RESEARCH AND INDUSTRY EXPERIENCE

CyberCARE Lab (USDOT Tier 1 UTC), University of Houston ***Houston, Texas***

Research Assistant; Advisor: Prof. Lu Gao *Aug 2025 - Present*

- Developed a human-centered GIS decision-support tool for TxDOT, integrating pavement conditions and 24,873 crash records across 18,843 roadway miles to identify roadway safety patterns and support agency decisions.
- Applied multimodal AI to analyze 25K+ construction-accident narratives and site images, evaluating 9 LLMs across 43 hazard categories (F1 up to 0.98) to understand contributing safety risks.

R.H. Sapat College of Engineering & Management ***Nashik, India***

Graduate Research Assistant; Advisor: Prof. Santosh Shinde *June 2024 – June 2025*

- Developed a skewness-based outlier elimination approach for the Pima Diabetes Dataset to improve diabetes prediction and classification performance.
- Applied statistical preprocessing, outlier detection, and machine learning techniques to the Pulsar Stars Dataset to improve data quality and classification accuracy.

RESEARCH PROJECTS

Gasoline Prices and Transit Ridership: Evidence from 50 U.S. Transit Agencies *2026*

- Analyzed a monthly panel of 50 U.S. transit agencies from January 2002 to March 2026 using fixed-effects models with gasoline prices, employment, fares, and post-COVID controls.
- Found that a 10% increase in real gasoline prices was associated with 2.1% higher total ridership, including increases of 1.3% for bus and 3.5% for rail.

AI Assistant for Knowledge Management and Workforce Training in State DOTs *2026*

- Developed a multi-agent retrieval-augmented generation system over 500+ State DOT reports, providing transportation professionals with source-grounded answers and access to institutional knowledge.
- Integrated vision-language models to interpret figures, charts, and tables and evaluated the system on 100 transportation domain queries, achieving Precision@3 of 1.00 and Recall@3 of 94.4%.

HONOURS AND AWARDS

- **Winner, Spring 2026 Coogs for Energy Hackathon** 🔗: First place among 50 competing teams; received a $6,000 team prize.
- **Selected Participant, NSF CyberSTAR 2026:** Selected for the NSF-funded CyberSTAR CyberTraining Workshop at George Mason University as one of 20 participants.
- **Presenter, 2nd CYBER-CARE Annual Symposium:** Presented *"LLM-Guided Hybrid Simulation for Airport Cyber-Resilience Assessment"* at Embry-Riddle Aeronautical University (ERAU).

SKILLS AND RELEVANT COURSEWORK

Transportation & Infrastructure Analytics: Transportation Safety, Pavement Condition Analysis, Crash Susceptibility Modeling, Transit Ridership Analysis, GIS, Panel Data Modeling

AI & Data-Driven Methods: Machine Learning, RAG, VLMs, Multi-Agent Systems, LLM Evaluation, Fixed-Effects Regression, Quantile Regression, Graph Neural Networks

Simulation & Research Tools: Python, PyTorch, SUMO, TraCI, JuPedSim, GIS, Git, REST APIs

Relevant Coursework: Data Analysis in Construction Management, Artificial Intelligence, Data Visualization, Digital Image Processing, Data Mining, Advanced Numerical Analysis

SERVICE AND VOLUNTEERING

- **IEEE Conference Reviewer:** *SMVITM 2026, ICDCECE 2026, INSTCon 2026, ICDSIS 2025.*
- **Student Committee Member, National Transportation Cybersecurity Competition, 2025 - 2026:** *Developed questions and supported USDOT CYBER-CARE competition coordination.*
- **Alumni Mentor, 2026 - Present:** *Mentor my bachelor's university undergraduate students through virtual career and leadership guidance sessions.*
- **Digital Literacy Volunteer, 2022 - 2024:** *Trained 30 older adults in digital literacy and smartphone applications.*
- **Rehabilitation Institute Volunteer, 2024:** *Supported activities for 100+ children, contributing 60+ volunteer hours.*